Akhil Gonugunta

Software Engineer | Hyderabad, India

Telecom Software Engineer | LTE/5G Packet Core | Python & MongoDB Specialist

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Profile Summary

Telecom Software Engineer with **5 years** of experience in **LTE/4G Core Network** development and testing. Expert in **Diameter (Gx, Rx, Gy, Sy)** protocols and **PCRF** policy logic. Proven track record in maintaining high-availability **MongoDB** clusters and automating system debugging using **Python**. Currently building scalable solutions for **Cisco Policy Suite (CPS)** with a focus on network efficiency and zero-downtime deployments.

Technical Skills

- **Core Domain:** LTE Architecture, PCRF, PCEF, DRA, 5G NSA Concepts, VoLTE Call Flows, WiFi (802.11 a/b/g/n/ac/ax), OpenWrt.
- **Protocols:** Diameter (RFC 6733), Gx, Rx, Gy, Sy, SCTP, TCP/IP, TLS/mTLS.
- **Databases:** MongoDB (Replica Sets, Sharding, Oplog Analysis).
- **Programming:** Python (Automation/Scripting), Java (Core Logic), Shell Scripting.
- **Tools & Infra:** Linux, Docker, Wireshark, Postman, VMware ESXi, Jira, Git.

Professional Experience

HCLTech (Client: Cisco)

Software Engineer | 2023 – Present

Key Project: Cisco Policy Suite (CPS) - Diameter Routing Agent (DRA)

- **DRA Logic Development:** Engineered and optimized **Diameter Routing Agent (DRA)** components to handle high-volume signaling traffic for **AT&T** (Client). Enhanced routing logic for Gx/Rx interfaces, improving session binding efficiency.

- **Database Engineering (MongoDB):** Developed and implemented high-availability MongoDB architectures across versions **5.0, 6.0, and 7.0**, executing critical zero-downtime migrations to ensure seamless production scalability.
- **Network Security:** Implemented **Mutual TLS (mTLS)** encryption for DRA signaling, ensuring end-to-end data confidentiality and compliance with carrier-grade security standards.
- **Automation & Tooling:** Developed **Python automation suites** to parse multi-gigabyte server logs and automate Health Checks, reducing Root Cause Analysis (RCA) turnaround time.
- **Infrastructure Management:** Managed **Dockerized microservices** and Linux-based servers. Created custom scripts for software deployment, scaling, and system recovery.
- **Documentation:** Authored High-Level Designs (HLD), Low-Level Designs (LLD), and Method of Procedures (MoPs) for complex network upgrades and feature activations.

HCLTech (Client: Rakuten)

Software Test Engineer (Protocol Testing) | 2021 – 2022

Key Project: Smart WiFi & Self-Organizing Networks (SON)

- **Protocol Validation:** Conducted rigorous testing of **802.11ax (WiFi 6)** protocols and **Self-Organizing Network (SON)** features for Enterprise and Outdoor Access Points.
- **Log Analysis:** Performed deep-dive analysis of packet captures using **Wireshark** to debug connectivity drops and handover failures, collaborating directly with the development team to fix critical firmware bugs.
- **Test Automation:** Coded and deployed wireless test environments to simulate complex network topologies, verifying the stability of **Wireless LAN Controllers (WLC)**.
- **Quality Assurance:** Designed comprehensive Test Plans and executed regression suites for WebUI and firmware stability, ensuring 100% test coverage before release.

Education

Bachelor of Technology (B.Tech) – Electronics & Communication Engineering

QIS College of Engineering and Technology, Ongole | 2019 | CGPA: 8.2

Declaration

I hereby Declare that all the above-mentioned details provided by me are true up to my knowledge.

Gonugunta Akhil